

Python

Basic Python

Python Fundamentals

- Intro Python
- Understanding variables and data types (int, float, string, boolean)
- Input and Output
- Casting (converting between data types)
- Mathematical and logical operators
- Conditionals (if, else, elif)
- Loops
 - for
 - Nested Loops
 - while
 - Infinite Loop
 - continue & break
- String
 - Intro String
 - String Methods
 - upper()
 - lower()
 - replace()
 - split()
 - find()
 - strip()
- Functions (definition and calling functions)
- File handling (reading and writing text files)

Data Structures

- Lists
 - Methods
 - append()
 - extend()
 - insert()
 - remove()
 - pop()
 - clear()
 - index()
 - count()
 - sort()
 - reverse()
 - copy()
 - enumerate
 - Iterating through lists with loops
 - Zip
 - List Comprehension
 - Reverse Iteration
- dictionaries
 - Methods
 - clear()
 - copy()
 - fromkeys(key, value)
 - get(key, default)
 - items()
 - keys()
 - pop(key, default)
 - popitem()
 - setdefault(key, default)
 - update()
 - values()
 - len()
 - min()
 - max()
 - sum()
 - in & not in
- tuples
 - Methods
 - max()
 - sum()
 - in & not in
 - packing and unpacking
- sets
 - Methods
 - Union
 - Intersection
 - Difference
- Basic operations on these structures (add, remove, search)
- Error Handling
 - Using try, except for error management

Numpy

- Introduction to NumPy
- NumPy Arrays
 - np.array()
 - np.zeros()
 - np.ones()
 - np.empty()
 - np.arange()
 - np.linspace()
 - shape
 - ndim
 - size
 - reshape()
 - ravel()
 - flatten()
- Data Types in NumPy
 - dtype
 - astype
 - reshape()
 - ravel()
 - flatten()
- Reshaping and Flattening
 - Basic Indexing
 - Slicing Arrays
 - Boolean Indexing
 - Fancy Indexing
- Indexing and Slicing
 - Basic Indexing
 - Slicing Arrays
 - Boolean Indexing
 - Fancy Indexing
- Array Operations
 - Element-wise Operations
 - Arithmetic operations
 - np.sqrt()
 - np.exp()
 - np.log()
 - Broadcasting
 - Vectorization
 - Matrix Multiplication
 - Transpose of Matrices
 - Determinant and Inverse of Matrices
 - np.linalg.det()
 - np.linalg.inv()
- Linear Algebra Operations
 - Descriptive Statistics
 - np.mean()
 - np.median()
 - np.var()
 - np.std()
 - np.sum()
 - Summation and Aggregation
 - np.cumsum()
 - np.prod()
 - Correlations and Covariance
 - np.corrcoef()
 - np.cov()
- Data Preprocessing
 - Normalization and Standardization
 - Standardization (Z-score normalization)
 - np.mean() and np.std()
 - Data Scaling
 - np.linalg.norm()
 - Max-min scaling with np.min() and np.max()
- Random Number Generation
 - Generating Random Data
 - np.random.rand()
 - np.random.randn()
 - Random Sampling
 - np.random.choice()
 - Seeding Seeds
 - np.random.seed()
 - Identifying Missing Values
 - np.isnan()
- Working with Missing Data
 - Identifying Missing Values
 - Imputing Missing Values

Pandas

- Introduction to Pandas
- Data Structures: Series and DataFrames
 - Series
 - DataFrames
 - DataFrame attributes
 - shape
 - columns
 - index
 - dtypes
 - iloc and iat
- Loading and Exporting Data
 - Reading Data
 - pd.read_csv()
 - pd.read_excel()
 - pd.read_sql()
 - Exporting Data
 - to_csv()
 - to_excel()
- Data Inspection and Summary
 - Viewing Data
 - head()
 - tail()
 - sample()
 - Summarizing Data
 - describe()
 - info()
 - isna()
 - isnull()
 - notna()
- Data Cleaning and Preprocessing
 - Handling Missing Data
 - dropna()
 - fillna()
 - Handling Duplicates
 - drop_duplicates()
 - drop_duplicates()
 - Renaming Columns and Indexes
 - rename()
 - Handling Outliers
 - Selecting Columns
- Selecting and Filtering Data
 - Selecting Rows
 - loc[]
 - iloc[]
 - Advanced Filtering
 - Modifying DataFrames
- Data Transformation
 - Data Type Conversion
 - Handling Categorical Data
 - get_dummies()
 - Scaling Data
 - StandardScaler()
 - MinMaxScaler()
- Merging, Joining, and Concatenation

Matplotlib & Seaborn

- Introduction to Data Visualization
- Line Plots
 - plt.plot()
 - plt.xlabel()
 - plt.ylabel()
 - plt.title()
- Scatter Plots
 - plt.scatter()
- Bar Charts
 - plt.bar()
 - plt.barh()
- Customizing Plots
 - Adding Legends
 - plt.legend()
 - Setting Axis Limits and Ticks
 - plt.xlim()
 - plt.ylim()
 - plt.xticks()
 - plt.yticks()
 - Subplots
 - plt.subplot()
 - plt.subplots()
 - plt.tight_layout()
- Annotating Plots
 - Adding Text Annotations
 - Highlighting Specific Data Points
- Saving Plots
 - plt.savefig()
- Visualizing Distributions
 - Histograms
 - plt.hist()
 - Box Plots
 - plt.boxplot()
 - Violin Plots (Seaborn Integration)
 - sns.violinplot()
 - Heatmaps
 - Heatmaps
 - Pair Plots (Pairwise Relationships)
- Seaborn Themes and Styles
 - darkgrid, whitegrid, ticks, etc.
 - sns.set_style()
 - sns.set_context()
- Seaborn Plots
 - Scatter Plots
 - Line Plots
- Distribution Visualization
 - Histograms and KDE (Kernel Density Estimation)
 - sns.histplot()
 - sns.kdeplot()
 - Joint Plots
 - sns.jointplot()
 - Box Plots and Violin Plots
 - sns.boxplot()
 - sns.violinplot()
- Categorical Data Visualization
 - Strip Plots
 - sns.stripplot()
 - Swarm Plots
 - sns.swarmplot()
 - Pair Plots
 - sns.pairplot()
 - Facet Grids
 - sns.FacetGrid()
- Color Palettes
- Plot Aesthetics
 - Adjusting Plot Size and Aspect Ratio
 - Multi-Plot Grids

Data Handling

- Text Data
 - Processing text files
 - Working with CSV and JSON data
- Regex (Regular Expressions)
 - pattern matching in strings (search, replace, etc.)

Tools & Frameworks

- Git & GitHub
- Flask & FastAPI
- Streamlit

Advanced Python

- Object-Oriented Programming (OOP)
- Multithreading and Parallel Processing
- Memory Management
- File and Path Handling
- APIs and Web Scraping
 - Fetching data from APIs using Requests
 - Web scraping with BeautifulSoup
- Handling Large Data
 - Dask

Projects

- File Management System
- Telegram Bot
- System Log Analysis
- System Automation Script
- File Download Tool from the Internet
- File Format Converter